



**POLITECNICO**  
MILANO 1863

**Academic Year 2024/2025**

**School of Civil, Environmental and Land Management Engineering**

**Courses offered by the programme**

**Civil Engineering (503)**

**Bachelor of Science degree (Ordinamento 270/04)**

**Milano Leonardo campus**

**Class of degree**

**L-7 - Civil and Environmental Engineering**

## Glossary

| Course type (Type) |                          |
|--------------------|--------------------------|
| I                  | Integrated Course        |
| M                  | Mono-Disciplinary Course |
| V                  | Final Examination        |

| Educational activities (Att Form) |                                   |
|-----------------------------------|-----------------------------------|
| C                                 | Similar or integrative activities |
| B                                 | Identifying activities            |
| A                                 | Basic activities                  |

| Innovative teaching                                                               |                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>The credits shown next to this symbol indicate the part of the course CFUs provided with Innovative teaching.<br/>These CFUs include:</p> <ul style="list-style-type: none"><li>- Subject taught jointly with companies or organizations</li><li>- Blended Learning &amp; Flipped Classroom</li><li>- Massive Open Online Courses (MOOC)</li><li>- Soft Skills</li></ul> |

| Semester (Sem) |                 |
|----------------|-----------------|
| 1              | First Semester  |
| 2              | Second Semester |
| A              | Annual course   |

| Sequences                                                                         |                       |
|-----------------------------------------------------------------------------------|-----------------------|
|  | Teachings in Sequence |

| Alphabetical group (Students from ... to ...)                    |  |
|------------------------------------------------------------------|--|
| Alphabetical groups are defined as FROM (included) TO (excluded) |  |

## 1st Year courses

### Track OAE - International track

Campus: Milano Leonardo

Track offered in English

| Code   | Act type | SSD                   | Course Title                                                                                                                                                                                          | Language | Type | Sem | CFU            | CFU Group |
|--------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|-----|----------------|-----------|
| 055716 | A        | MAT/03<br>MAT/05      | MATHEMATICAL ANALYSIS I AND GEOMETRY<br><i>Students form A to ZZZZ</i><br>- 055715 GEOMETRY - Frapporti Davide<br>- 055714 MATHEMATICAL ANALYSIS I - Piovano Paolo                                    | EN       | I    | 1   | 12.0           | 12.0      |
| 055718 | A        | CHIM/07               | CHEMISTRY<br><i>Students form A to ZZZZ - Sansotera Maurizio</i>                                                                                                                                      | EN       | M    | 1   | 6.0            | 6.0       |
| 055723 | B        | GEO/05                | ENGINEERING GEOLOGICAL SURVEY <sup>(a)</sup><br><i>Students form A to ZZZZ - Gattinoni Paola</i>                                                                                                      | EN       | M    | 1   | 6.0<br>[1.0 @] | 6.0       |
| 055721 | A        | FIS/01                | PHYSICS I AND PHYSICS IIA <sup>(b)</sup><br><i>Students form A to ZZZZ</i><br>- 055720 PHYSICS IIA - 2 Sem - Sordan Roman<br>- 055719 PHYSICS I - Khursheed Anjam                                     | EN       | I    | A   | 14.0           | 14.0      |
| 059047 | A,B      | ICAR/06<br>ING-INF/05 | FUNDAMENTALS OF COMPUTER SCIENCE AND TECHNICAL DRAWING<br><i>Students form A to ZZZZ</i><br>- 059046 TECHNICAL DRAWING - De Gaetani Carlo Iapige<br>- 059045 COMPUTER SCIENCE - Metelli Alberto Maria | EN       | I    | 2   | 10.0           | 10.0      |
| 055713 | A        | MAT/05                | MATHEMATICAL ANALYSIS II<br><i>Students form A to ZZZZ - Jaksic Vojkan</i>                                                                                                                            | EN       | M    | 2   | 8.0            | 8.0       |
| 055722 | C        | ING-IND/22            | CONSTRUCTION MATERIALS<br><i>Students form A to ZZZZ - Lollini Federica</i>                                                                                                                           | EN       | M    | 2   | 6.0            | 6.0       |

(a) d.i. = 1 CPS innovative teaching, Flipped classroom.

(b) Physics I annual teaching, Physics II A 2th semester.

### Track OAI - Percorso in italiano

Campus: Milano Leonardo

Track offered in Italian

| Code   | Act type | SSD                                      | Course Title                                                                                                                                                                            | Language | Type | Sem | CFU            | CFU Group |
|--------|----------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------|-----|----------------|-----------|
| 095730 | A        | MAT/03<br>MAT/05                         | MATHEMATICAL ANALYSIS 1.ST AND GEOMETRY<br><i>Students form A to ZZZZ</i><br>- 095729 GEOMETRY - Lella Paolo<br>- 095728 MATHEMATICAL ANALYSIS 1.ST - Potrich Norman                    | IT       | I    | 1   | 12.0           | 12.0      |
| 097247 | A        | CHIM/07                                  | CHEMISTRY A<br><i>Students form A to ZZZZ - Di Pietro Maria Enrica</i>                                                                                                                  | IT       | M    | 1   | 6.0            | 6.0       |
| 097264 | A        | FIS/01                                   | PHYSICS 1ST AND PHYSICS IIA<br><i>Students form A to ZZZZ</i><br>- 097263 PHYSICS IIA - 2 Sem - Nisoli Mauro<br>- 097261 PHYSICS I - Brambilla Alberto Luigi                            | IT       | I    | A   | 14.0           | 14.0      |
| 059050 | A,B      | ICAR/06<br>ING-INF/05                    | FUNDAMENTALS OF COMPUTER SCIENCE AND TECHNICAL DRAWING<br><i>Students form A to ZZZZ</i><br>- 059049 TECHNICAL DRAWING - Rossi Lorenzo<br>- 059048 COMPUTER SCIENCE - Restelli Marcello | IT       | I    | 2   | 10.0           | 10.0      |
| 055712 | A        | MAT/05                                   | MATHEMATICAL ANALYSIS 2<br><i>Students form A to ZZZZ - Cavagnari Giulia</i>                                                                                                            | IT       | M    | 2   | 8.0            | 8.0       |
| 095731 | C        | ING-IND/22                               | MATERIALS FOR CIVIL ENGINEERING<br><i>Students form A to ZZZZ - Carsana Maddalena</i>                                                                                                   | IT       | M    | 2   | 6.0            | 6.0       |
| 061966 | B,C      | ICAR/02<br>ICAR/04<br>ICAR/07<br>ICAR/09 | INTRODUCTION TO CIVIL ENGINEERING <sup>(a)</sup><br><i>Students form A to ZZZZ - Smerzini Chiara</i>                                                                                    | IT       | M    | 1   | 3.0<br>[3.0 @] | 3.0       |

(a) d.i.= 3CFU Soft Skills.

## 2nd Year courses

### Track OAE - International track

Campus: Milano Leonardo

Track offered in English

| Code   | Act type | SSD        | Course Title                                                                                              | Language | Type | Sem | CFU             | CFU Group |
|--------|----------|------------|-----------------------------------------------------------------------------------------------------------|----------|------|-----|-----------------|-----------|
| 059052 | A        | MAT/05     | DIFFERENTIAL EQUATIONS <sup>(a)</sup><br><i>Students form A to ZZZZ - Abatangelo Laura</i>                | EN       | M    | 1   | 8.0<br>[1.0 @]  | 8.0       |
| 056794 | A        | MAT/07     | RATIONAL MECHANICS<br><i>Students form A to ZZZZ - Riccobelli Davide</i>                                  | EN       | M    | 1   | 10.0            | 10.0      |
| 056793 | B        | ICAR/06    | SURVEYING AND DATA PROCESSING<br><i>Students form A to ZZZZ - Barzaghi Riccardo</i>                       | EN       | M    | 1   | 12.0            | 12.0      |
| 056795 | B        | ICAR/01    | HYDRAULICS<br><i>Students form A to ZZZZ - Messa Gianandrea Vittorio</i>                                  | EN       | M    | 2   | 10.0            | 10.0      |
| 056797 | B        | ICAR/08    | STRUCTURAL MECHANICS <sup>(b)</sup><br><i>Students form A to ZZZZ - Taliercio Alberto</i>                 | EN       | M    | 2   | 12.0<br>[1.0 @] | 12.0      |
| 056798 | C        | ING-IND/35 | PROJECT MANAGEMENT: PRINCIPLES & TOOLS<br><i>Students form A to ZZZZ - Sainati Tristano</i>               | EN       | M    | 2   | 6.0             | 6.0       |
| 097302 | C        | ING-IND/10 | THERMODYNAMICS AND HEAT TRANSFER (CIVIL ENGINEERING)<br><i>Students form A to ZZZZ - Ferrando Martina</i> | IT       | M    | 2   | 6.0             |           |

(a) d.i. = 1 CPS innovative teaching, MOOC.

(b) d.i. = 1 CPS innovative teaching, Flipped classroom.

### Track OAI - Percorso in italiano

Campus: Milano Leonardo

Track offered in Italian

| Code   | Act type | SSD        | Course Title                                                                                              | Language | Type | Sem | CFU             | CFU Group |
|--------|----------|------------|-----------------------------------------------------------------------------------------------------------|----------|------|-----|-----------------|-----------|
| 059062 | A        | MAT/05     | DIFFERENTIAL EQUATIONS <sup>(a)</sup><br><i>Students form A to ZZZZ - Garrione Maurizio</i>               | IT       | M    | 1   | 8.0<br>[1.0 @]  | 8.0       |
| 085644 | A        | MAT/07     | RATIONAL MECHANICS<br><i>Students form A to ZZZZ - Fermi Davide</i>                                       | IT       | M    | 1   | 10.0            | 10.0      |
| 056792 | B        | ICAR/06    | DATA PROCESSING (SURVEYING)<br><i>Students form A to ZZZZ - Albertella Alberta</i>                        | IT       | M    | 1   | 10.0            | 10.0      |
| 085649 | B        | ICAR/01    | HYDRAULICS<br><i>Students form A to ZZZZ - Mambretti Stefano</i>                                          | IT       | M    | 2   | 10.0            | 10.0      |
| 056796 | B        | ICAR/08    | STRUCTURAL MECHANICS <sup>(b)</sup><br><i>Students form A to ZZZZ - Perego Umberto</i>                    | IT       | M    | 2   | 12.0<br>[1.0 @] | 12.0      |
| 097302 | C        | ING-IND/10 | THERMODYNAMICS AND HEAT TRANSFER (CIVIL ENGINEERING)<br><i>Students form A to ZZZZ - Ferrando Martina</i> | IT       | M    | 2   | 6.0             | 6.0       |
| 054249 | B        | GEO/05     | ENGINEERING GEOLOGICAL SURVEY <sup>(c)</sup><br><i>Students form A to ZZZZ - Papini Monica</i>            | IT       | M    | 2   | 6.0<br>[1.0 @]  | 6.0       |

(a) d.i. = 1 CPS innovative teaching, MOOC.

(b) d.i. = 1 CPS innovative teaching, Flipped classroom.

(c) d.i. = 1 CPS innovative teaching, Flipped Classroom

## 3rd Year courses

### Track OAE - International track

Campus: Milano Leonardo

Track offered in English

| Code   | Act type | SSD                   | Course Title                                                                                                            | Language | Type | Sem | CFU            | CFU Group |
|--------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|----------|------|-----|----------------|-----------|
| 057994 | B        | ICAR/09               | STRUCTURAL DESIGN<br><i>Students form A to ZZZZ - Di Luzio Giovanni</i>                                                 | EN       | M    | A   | 10.0           | 12.0      |
| 057995 | --       | --                    | FINAL EXAM (CIV-ICAR/09) <sup>(a)</sup><br><i>Students form A to ZZZZ - Di Luzio Giovanni</i>                           | EN       | V    | A   | 2.0<br>[2.0 @] |           |
| 058101 | B        | ICAR/07               | GEOTECHNICS<br><i>Students form A to ZZZZ - Jommi Cristina</i>                                                          | EN       | M    | 1   | 10.0           | 10.0      |
| 057999 | B        | ICAR/02               | HYDRAULIC ENGINEERING<br><i>Students form A to ZZZZ - Sanfilippo Umberto</i>                                            | EN       | M    | 1   | 9.0            | 10.0      |
| 058000 | --       | --                    | FINAL EXAM (CIV-ICAR/02) <sup>(b)</sup><br><i>Students form A to ZZZZ - Sanfilippo Umberto</i>                          | EN       | V    | 1   | 1.0<br>[1.0 @] |           |
| 058001 | B,C      | ICAR/04               | CONSTRUCTION OF ROADS, RAILWAYS AND AIRPORTS<br><i>Students form A to ZZZZ - Ketabdari Misagh</i>                       | EN       | M    | 2   | 10.0           | 10.0      |
| 058002 | B        | GEO/11                | NON DESTRUCTIVE TESTING<br><i>Students form A to ZZZZ - Hojat Azadeh</i>                                                | EN       | M    | 2   | 6.0            | 6.0       |
| 058003 | C        | ING-IND/12            | STRUCTURAL MONITORING<br><i>Students form A to ZZZZ - Cigada Alfredo</i>                                                | EN       | M    | 2   | 6.0            | 6.0       |
| 058004 | C        | ICAR/03               | ENVIRONMENTAL ENGINEERING<br><i>Students form A to ZZZZ - Azzellino Arianna</i>                                         | EN       | M    | 1   | 6.0            | 6.0       |
| 058005 | C        | ICAR/22<br>ING-IND/35 | PRINCIPLES OF ECONOMICS AND ECONOMIC EVALUATION OF PROJECTS<br><i>Students form A to ZZZZ - Bassi Andrea</i>            | IT       | M    | 2   | 6.0            |           |
| 088502 | C        | ING-IND/22            | DURABILITY OF MATERIALS AND REPAIR TECHNOLOGIES OF STRUCTURES<br><i>Students form A to ZZZZ - Gastaldi Matteo Maria</i> | IT       | M    | 1   | 6.0            |           |
| 088508 | C        | ING-IND/31            | ELECTRICAL ENGINEERING<br><i>Students form A to ZZZZ - Negri Simone</i>                                                 | IT       | M    | 2   | 6.0            |           |
| 088510 | B,C      | ICAR/04               | CONSTRUCTION OF INFRASTRUCTURE FACILITIES<br><i>Students form A to ZZZZ - Oliva Massimo</i>                             | IT       | M    | 2   | 6.0            |           |

(a) d.i.= 2CFU Flipped classroom

(b) d.i.= 1CFU Flipped classroom

### Track OAI - Percorso in italiano

Campus: Milano Leonardo

Track offered in Italian

| Code   | Act type | SSD                   | Course Title                                                                                                            | Language | Type | Sem | CFU            | CFU Group |
|--------|----------|-----------------------|-------------------------------------------------------------------------------------------------------------------------|----------|------|-----|----------------|-----------|
| 052334 | B        | ICAR/09               | STRUCTURAL ANALYSIS AND DESIGN<br><i>Students form A to ZZZZ - Di Prisco Marco</i>                                      | IT       | M    | A   | 10.0           | 12.0      |
| 054257 | --       | --                    | FINAL DEGREE TEST (2 CPS - ICAR/09) <sup>(a)</sup><br><i>Students form A to ZZZZ - Di Prisco Marco</i>                  | IT       | V    | A   | 2.0<br>[2.0 @] |           |
| 058100 | B        | ICAR/07               | GEOTECHNICS<br><i>Students form A to ZZZZ - Di Prisco Claudio Giulio Mario</i>                                          | IT       | M    | 1   | 10.0           | 10.0      |
| 052336 | B        | ICAR/02               | HYDRAULIC ENGINEERING<br><i>Students form A to ZZZZ - Becciu Gianfranco</i>                                             | IT       | M    | 1   | 9.0            | 10.0      |
| 054258 | --       | --                    | FINAL DEGREE TEST (1 CPS - ICAR/02) <sup>(b)</sup><br><i>Students form A to ZZZZ - Becciu Gianfranco</i>                | IT       | V    | 1   | 1.0<br>[1.0 @] |           |
| 099229 | B,C      | ICAR/04               | CONSTRUCTION OF ROADS, RAILWAYS AND AIRPORTS<br><i>Students form A to ZZZZ - Crispino Maurizio</i>                      | IT       | M    | 2   | 10.0           | 10.0      |
| 052338 | C        | ICAR/22<br>ING-IND/35 | PRINCIPLES OF ECONOMICS, COST AND ESTIMATING ENGINEERING<br><i>Students form A to ZZZZ - Bassi Andrea</i>               | IT       | M    | 2   | 5.0            | 5.0       |
| 058004 | C        | ICAR/03               | ENVIRONMENTAL ENGINEERING<br><i>Students form A to ZZZZ - Azzellino Arianna</i>                                         | EN       | M    | 1   | 6.0            | 12.0      |
| 088502 | C        | ING-IND/22            | DURABILITY OF MATERIALS AND REPAIR TECHNOLOGIES OF STRUCTURES<br><i>Students form A to ZZZZ - Gastaldi Matteo Maria</i> | IT       | M    | 1   | 6.0            |           |
| 088508 | C        | ING-IND/31            | ELECTRICAL ENGINEERING<br><i>Students form A to ZZZZ - Negri Simone</i>                                                 | IT       | M    | 2   | 6.0            |           |

|        |     |                                          |                                                                                                      |    |   |   |             |  |
|--------|-----|------------------------------------------|------------------------------------------------------------------------------------------------------|----|---|---|-------------|--|
| 088510 | B,C | ICAR/04                                  | CONSTRUCTION OF INFRASTRUCTURE FACILITIES<br><i>Students form A to ZZZZ - Oliva Massimo</i>          | IT | M | 2 | 6.0         |  |
| 056798 | C   | ING-IND/35                               | PROJECT MANAGEMENT: PRINCIPLES & TOOLS<br><i>Students form A to ZZZZ - Sainati Tristano</i>          | EN | M | 2 | 6.0         |  |
| 061966 | B,C | ICAR/02<br>ICAR/04<br>ICAR/07<br>ICAR/09 | INTRODUCTION TO CIVIL ENGINEERING <sup>(c)</sup><br><i>Students form A to ZZZZ - Smerzini Chiara</i> | IT | M | 1 | 3.0<br>[3.0 |  |

(b) d.i.= 1CFU Flipped classroom

(c) d.i.= 3CFU Soft Skills.